

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I R. Rupasree (192H79182) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Rupasree

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RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1] Smart Home Automation System Using Intelligent Agents :-

Introduction :-

A Smart Home Automation System uses AI to automatically control lighting, temperature & security based on user behavior and environmental conditions. Intelligent agent helps system make decision and improve performance over time.

1) Simple Reflex Agent :-

It acts only on current sensor input using predefined IF-THEN rules.

Eg:- If motion is detected $\rightarrow$ Turn ON light
If smoke is detected $\rightarrow$ Activate fire alarm.

Advantage :-

- > Fast Response
- > Easy to Implement

Disadvantage :-

- > No memory
- > Cannot learn user preference.

2) Model-Based Agent :-

It maintains an internal model of environment and remembers previous states.

Eg:- If TV is ON but no motion is detected, it assume someone is still in room & keep lights ON.

Advantage :-

- > Uses Memory
- > Better decision making

Disadvantage :-

- > More complex
- > Requires more storage

3) Goal-Based Agent :-

It selects actions to achieve a specific goal.

Eg:- If room temperature rises above 30°C , it turn ON air conditioner to reach desired temperature.

Advantages:-

- > Intelligent planning
- > Achieve long-term goal

Disadvantage:-

- > slower than reflex agent
- > Requires planning.

4) Utility-Based Agent:-

It choose the action that provide highest overall benefit by considering comfort, energy usage and security.

Eg:- Instead of turning on AC, it may open windows if outside weather is pleasant to save electricity.

Advantage:-

- > optimizes comfort and energy efficiency
- > Learns user preference.

Disadvantages:-

- > More complex
- > Needs utility calculations

A Hybrid Agent combining Simple Reflex, Model-Based and Goal Based agents

- > provides quick responses to emergencies
- > Learn user preference
- > optimizes energy consumption

It combines fast decision making, learning ability, planning and optimization to provide comfort, energy efficiency.

2. Robot Navigation in an Unknown Maze Using UCS and IDS:-

A robot placed in unknown maze must find exit without knowing layout. Uninformed search algorithm such as UCS and IDS helps the robot explore maze and reach goal.

UCS - Uniform Cost Search:-

It expands node with lowest path cost first and always find least-cost path.

Eg → If one path costs 5 and another costs 8,

UCS choose the path with cost 5.

Advantages:-

- > Finds optimal path & Complete Algorithm.
- > Works well when movement costs vary.

Disadvantages:-

- > Uses large Memory
- > Slow for large search spaces

Time Complexity :- $O(b^{1+c^*/\epsilon})$

Space Complexity :- $O(b^{1+c^*/\epsilon})$

Iterative Deepening Search (IDS):-

IDS repeatedly performs Depth-limited Search with increasing depth until goal is found.

Eg:- Robot Searches depth 1, then depth 2 then depth 3 until it reaches the exit

Advantages:-

- > low memory usage
- > Complete algorithm

Disadvantages:-

- > Repeat searches
- > Slower due to repeated exploration.

Time complexity :- $O(b^d)$

Space complexity :- $O(b^d)$

Intelligent Agents:-

Simple Reflex Agent

- Acts only on current sensor input
- May get stuck in loops

Model-Based Agent

- Stores Visited locations
- Avoids revisiting same path

Goal-Based Agent

- Plans actions reaches the exit
- Uses search algorithm effectively.

Utility-Based Agent

- Selects best path considering distance, energy and safety

Most effective solution is model-Based Goal Based Agent with Uniform Cost Search (UCS).

- Remembers explored path
- plans effectively toward exit
- Finds lowest cost route

Uniform cost search is suitable for mazes with different movement costs, which IDS is useful when memory is limited. A model based & goal agent combined UCS provides best performance for efficient, reliable and intelligent robot navigation.